Protective Relays

Overcurrent Relay :

Formulae in O.C Relay :

(1) Pickup current (or) plug settings in Amps.

I_{PK} = (% PSB setting value) × C.T secondary current

(2) Plug setting multiplier (PSM)

$$PSM = \frac{I_f}{I_{PK} \times CT\ Ratio}$$

$$I_f = PSM \times I_{PK} \times CT\ Ratio = PSM \times PSB\ setting \times I_s \times \frac{I_p}{I_s}$$

Assume PSB setting = 100 % = 1 ; $I_p = I_L$ = Full load current

$$I_f = PSM \times I_L$$

$I_f \propto PSM$; PSM = Multiplication factor of full load current.

(3) Actual Time of operating of Relay (ATO)

ATO = TMS × Time calculated from IDMT graph or Table

$$ATO = \frac{0.14 \times TMS}{(PSM)^{0.02} - 1}\ ; \quad TMS = \text{Time Multiplier setting}$$

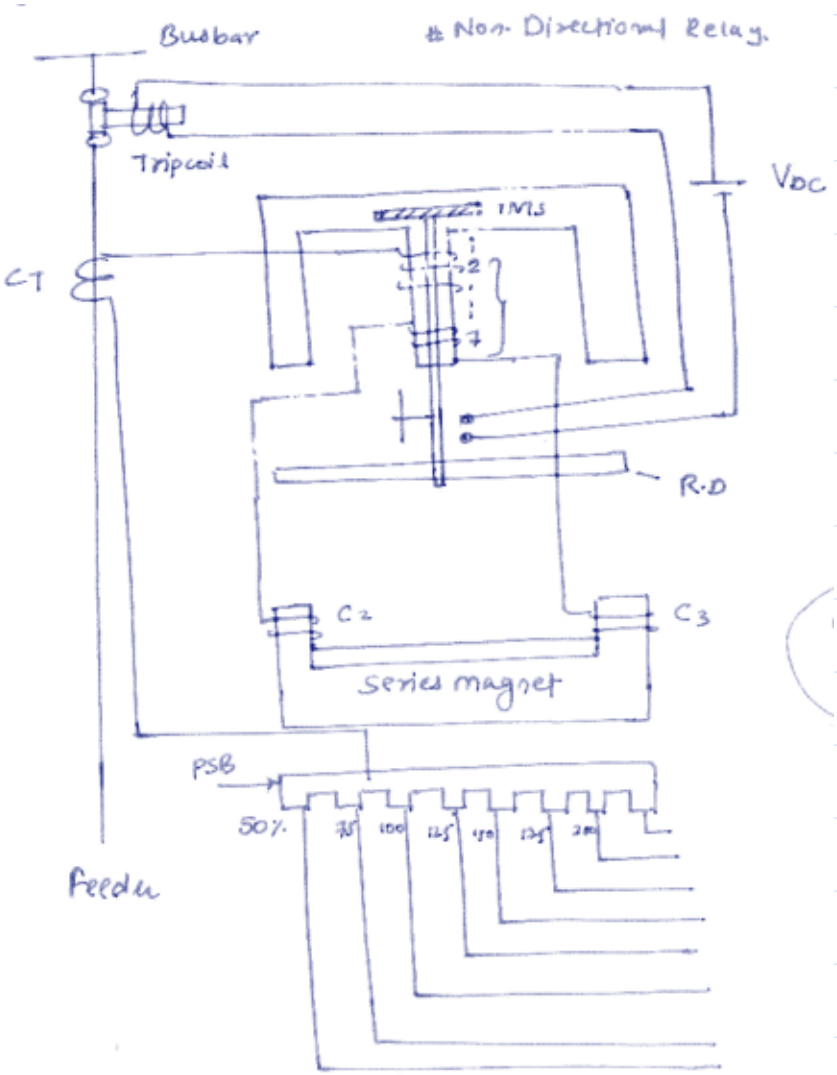
$$CT\ Ratio \text{ is not given : } CT\ Ratio = \frac{I_p}{I_s} = \frac{I_L}{I_{PK}} = \frac{\text{Full load current}}{\text{pickup current}}$$

(4) $T_s = T_p + T_d$

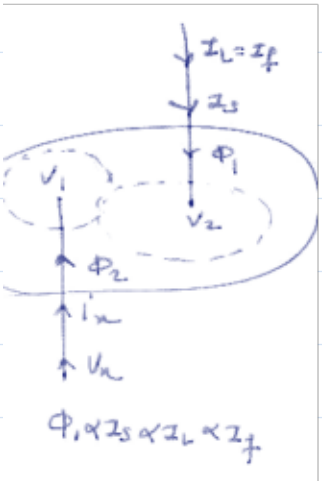
T_s = ATO of Backup (OR) Secondary Relay ; T_p = ATO of primary Relay ; T_d = Time Delay in Backup and primary Relay.

PSB -> Plug settings bridge ; R.D -> Rotating Disc ; S -> control spring ; C_1, C_2, C_3 -> copper coils ; TMS -> Time multiplier settings ; TC -> Trip contacts ; P -> plunger.

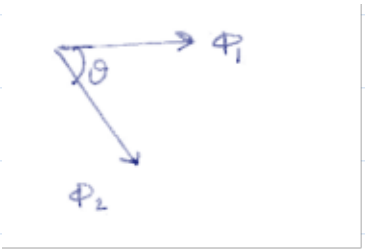
Non-Directional Relay.



Non-directional overcurrent induction relay: busbar/feeder CT feeding relay coil with taps through plug setting bridge PSB (50, 75, 100, 125, 150, 175, 200 %), series magnet with copper coils C2, C3, rotating disc R.D, TMS adjustment, trip contacts and trip coil with Vdc



Flux phasors: I_L = I_f, I_s, flux phi1 and phi2, V1, V2; phi1 proportional to Is proportional to IL proportional to If

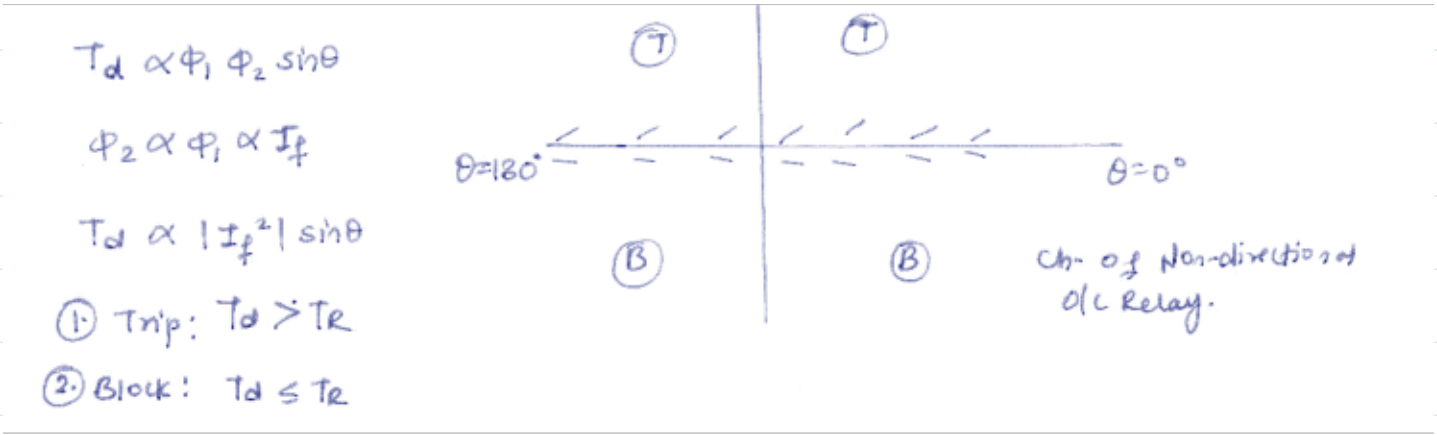


Phasors phi1 and phi2 with angle theta

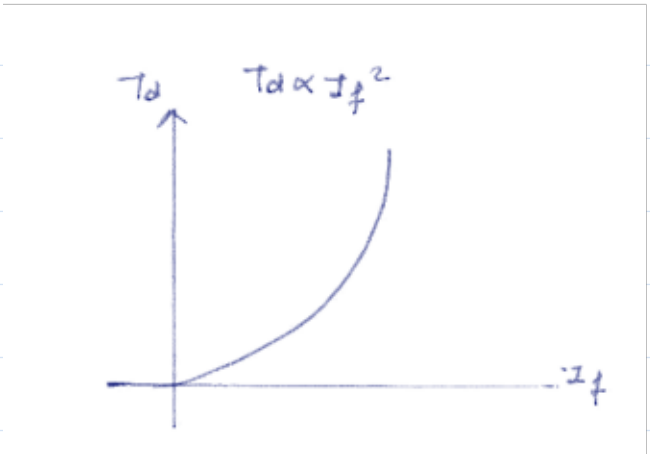
Trip : (T) , Block : (B) ; # Symbols for Non-directional O/C Relay (double arrow with a cross on the line)

$T_d \propto \phi_1 \phi_2 \sin\theta$; $\phi_2 \propto \phi_1 \propto I_f$; $T_d \propto |I_f^2| \sin\theta$

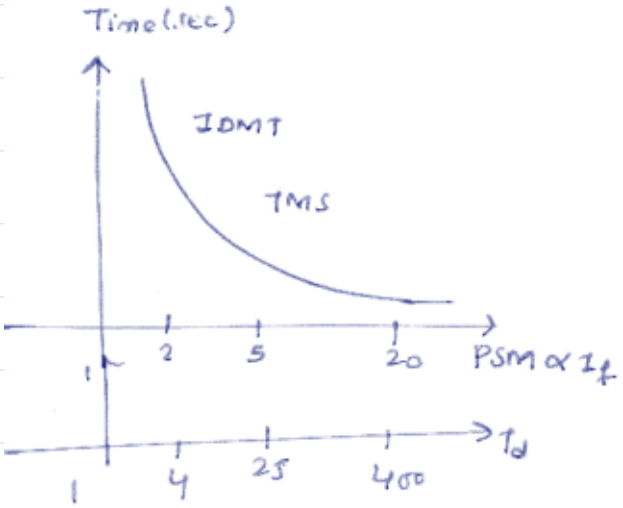
(1) Trip : $T_d > T_R$; (2) Block : $T_d \leq T_R$



Characteristic of non-directional O/C relay: Trip (T) zones above the line and Block (B) zones below, theta = 180 degrees to theta = 0 degrees



Td against If, parabolic: Td proportional to If squared



IDMT: time (sec) against PSM proportional to If (1, 2, 5, 20) with TMS; second axis Td (1, 4, 25, 400)

Time Grading OR TMS :

For maintaining proper time gradation between primary and backup relays, time grading is used. In this, the distance between plunger and trip contacts is adjusted to maintain proper time delays between the relays.

Lockout or Stuck Breaker :

If sufficient amount of gas pressure is not available in the Bkr, Bkr is kept in closed position even if fault is present. This is called Lockout or stuck breaker. During this time backup relays and breaker will isolate the fault with time delay.

Current grading or PSB :

* Over current relays are used over wide range of current settings by using PSB.
* Normal range is from 50 to 200 % in steps of 25 %.
* In case of Earth Fault relay this setting is from 20 % to 80 % in steps of 10 %.

In case of 3-ϕ system possible S.C faults are 11 faults; for protection of the systems from these faults, minimum no. of relays required are 3, i.e. 2 O/C and 1 E/F relays.

Eg: PSM / 2 / 4 / 5 / 8 / 10 / 20 (TMS = 1)

Oper. Time | 10 | 5 | 4 | 3 | 2.8 | 2.4

Parameter | R_1 | R_2

I_{PK} | 5 | $5 \times 1.25 = 6.25$

$PSM = \frac{I_f}{I_{PK} \times CT\ ratio}$ | $\frac{5000}{5 \times 1000/5} = 5$ | $\frac{5000}{6.25 \times 1000/5} = 4$

T for TMS = 1 | 4 | 5

TMS | 0.3 | $\frac{1.7}{5} = 0.34$

T sec | $4 \times 0.3 = 1.2 = T_p$ | $T_p + T_d = 1.2 + 0.5 = 1.7$

Eg:

$$t_{op} = \frac{0.14 \times 0.1}{(20)^{0.02} - 1} = 0.2278 \text{ sec}; \qquad PSM = \frac{2000}{5 \times \frac{100}{5}} = 20$$

$$CT\ Ratio = \frac{Full\ load\ current}{Pickup\ current} = \frac{100}{5}$$

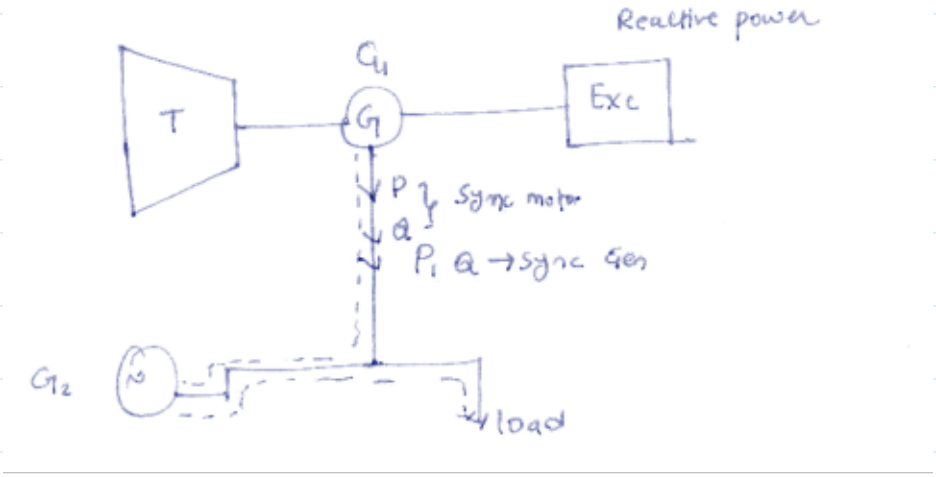
Non-directional O/C Relay :

Identifies the fault both in the forward and backward directions for isolation of faulty sections effectively. This is practically used in source end for isolation of faults on line side and source side.

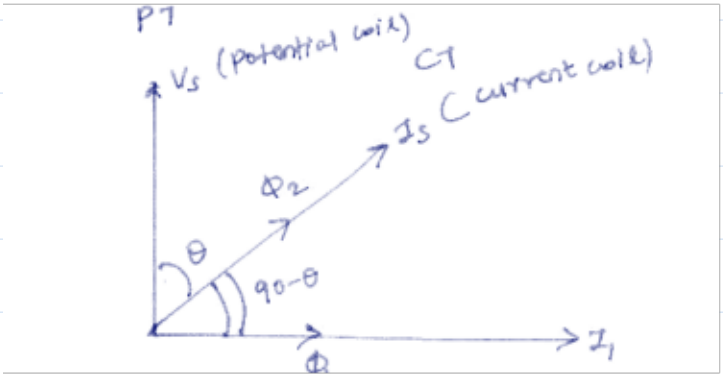
Directional or Reverse power OR watt-metric Relay :

Directional Relay is working on the basis of power flow, which is used practically for protection of the generator from failure of prime mover (Turbine).

Application : Failure of Turbine (Prime mover) of Generator.



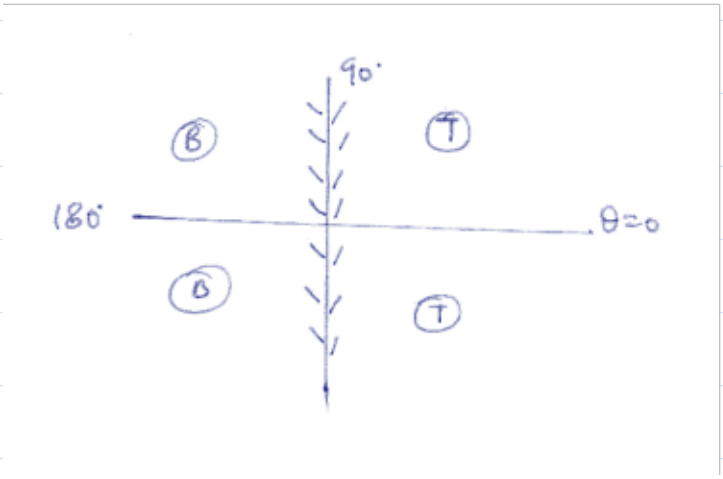
Turbine T driving generator G1 with exciter (reactive power); P, Q flow into G1 acting as a synchronous motor, from generator G2 and load on the common bus



Phasors: Vs (potential coil, PT), Is (current coil, CT), flux phi2 along Is, phi1 along I1, angle theta from Vs and 90 - theta

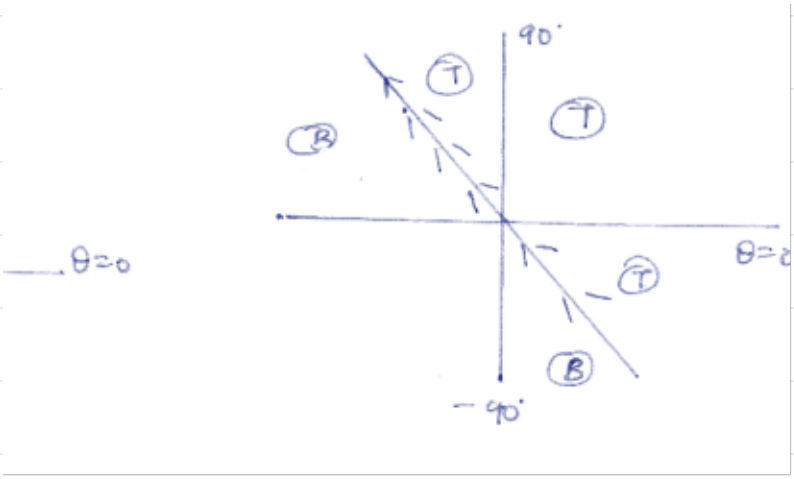
$$T_d \propto \phi_1 \phi_2 \sin(90-\theta)\ ;\ \phi_1 \propto V_s\ ,\ \phi_2 \propto I_s$$

$$\boxed{T_d \propto V_s I_s \cos\theta = \text{power}}$$



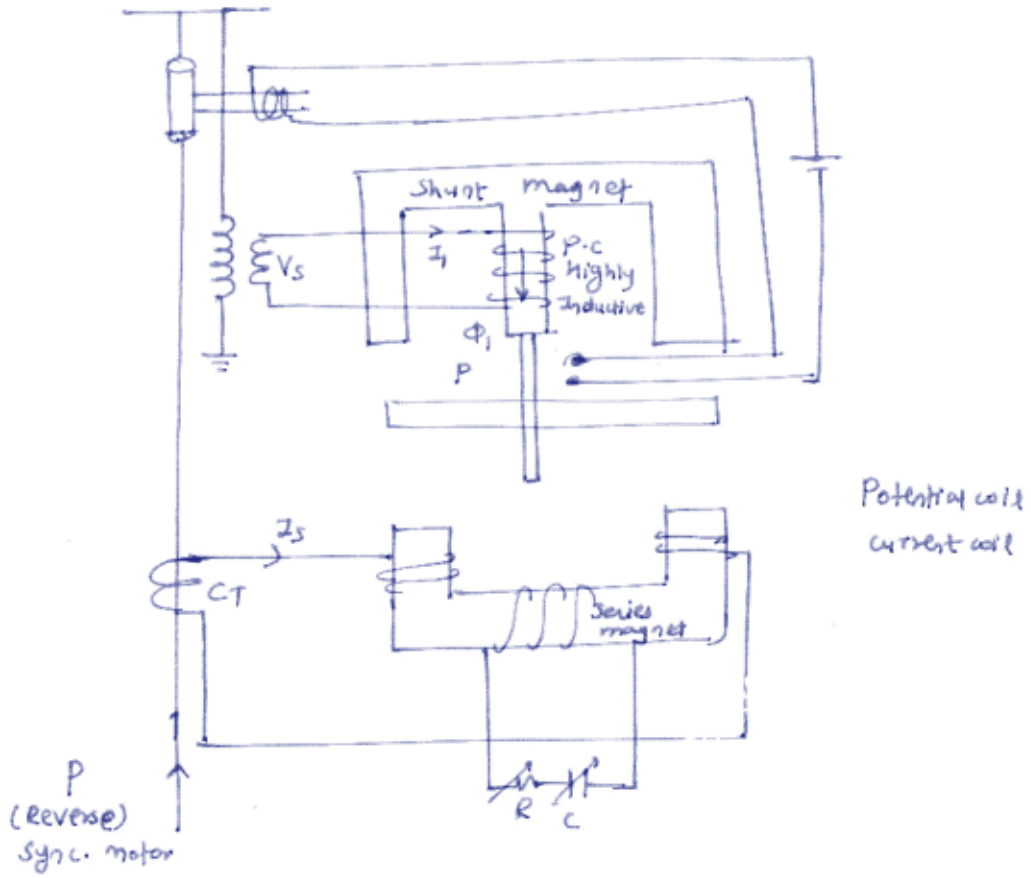
Directional relay characteristic: trip (T) on right of the 90-degree line, block (B) on left, theta = 0 to 180

Practical : $T_d \propto V_s I_s \cos(\theta - \tau)$; τ = RC = Max torque adjustment = Angle (20 to 60 degrees)



Practical characteristic: trip/block boundary line tilted by tau from the 90 degree axis

Directional Relay :



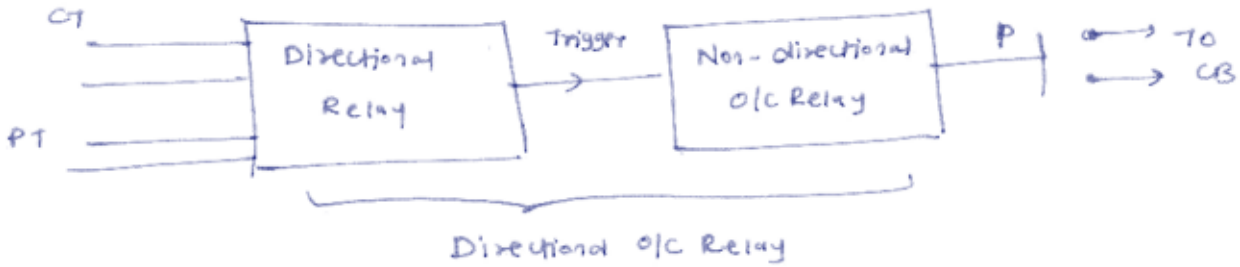
Directional relay: shunt magnet with potential coil (Vs, highly inductive) and series magnet with current coil (Is from CT), R and C network, plunger P and trip contacts, bus feeder with reverse power P from synchronous motor

(1) Directional O/C Relay (DOC) internally has 2 relays, in which one is directional and other is non-directional O/C Relay.

(2) If direction of the power is reverse, then directional relay will trigger the non-directional OC relay and this relay produces Trip signal if the magnitude of the fault current is more than relay PSB set values.

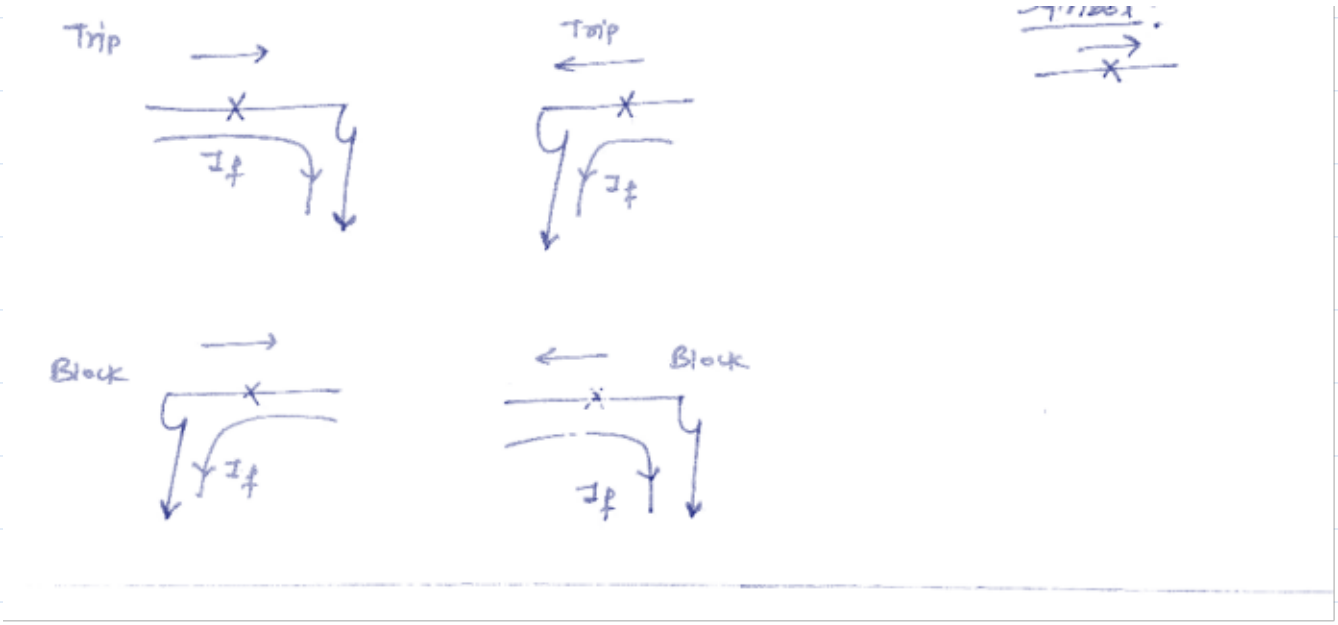
O/C -> Non-directional over current Relay ; # D.O.C -> Directional Over current Relay.

* If relay arrow direction and fault current direction are same then only relay will produce trip signal. In opposite direction relay block.



CT and PT inputs to Directional relay which triggers the Non-directional O/C relay, whose output P goes to the trip circuit of the CB; together they form the Directional O/C relay

Symbol : an arrow with a cross on the line (directional).



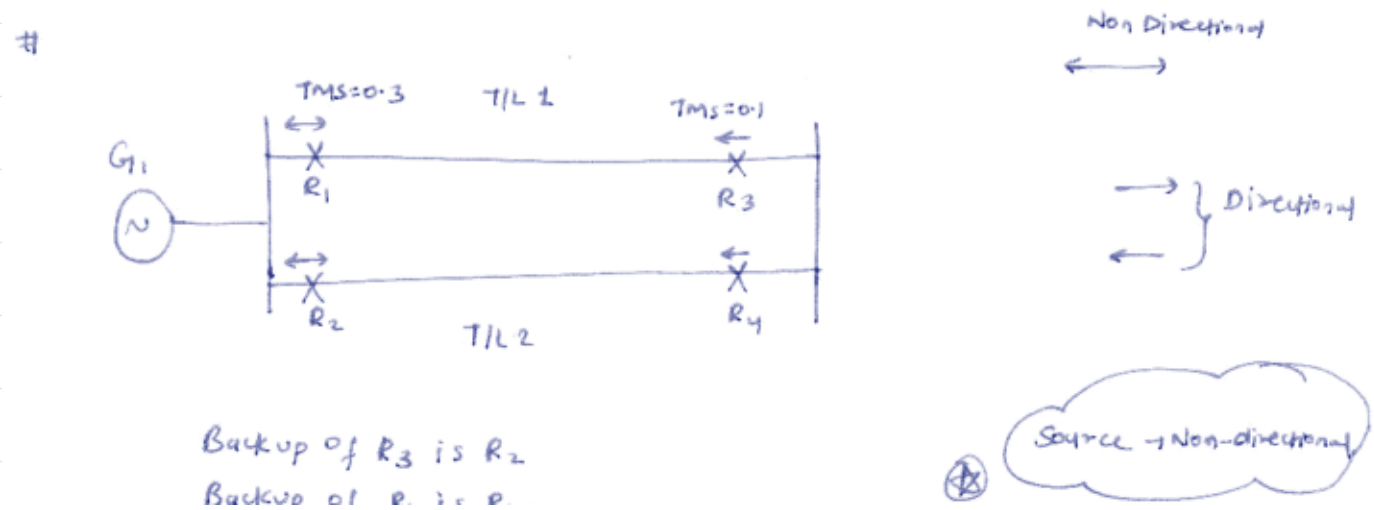
Trip when relay arrow and fault current If point the same way; Block when they are opposite

Application of Non-directional O/C and Directional O/C Relay :

(1) Protection of parallel feeder : In the parallel operation of T/L, over current relays are kept at the source side and DOC relays are kept on the other side of the line.

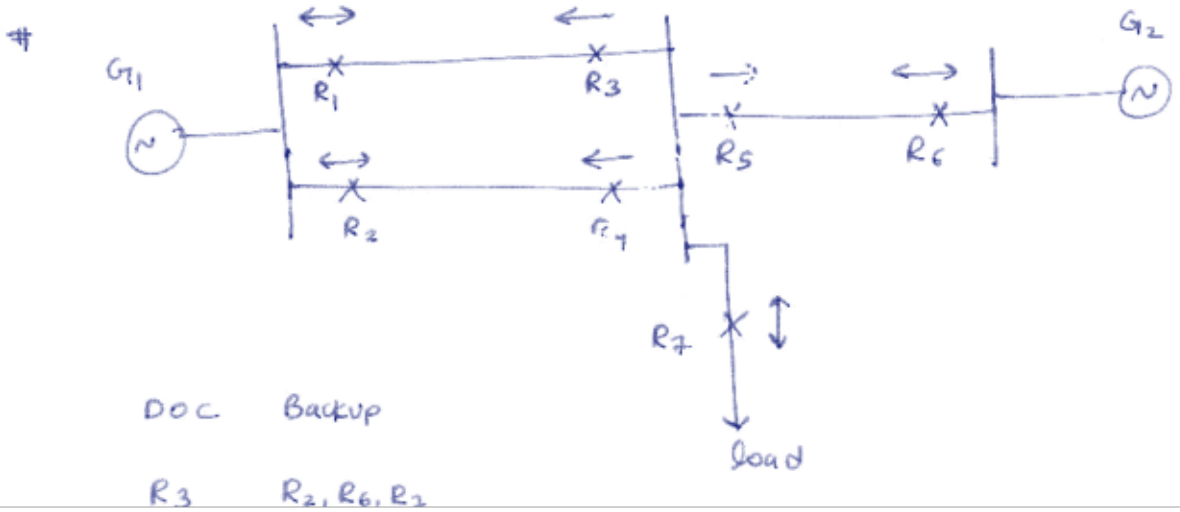
Proper time grading is maintained between the relays for avoiding mal-operation of the relays.

If any relay is producing trip signal without any fault, it is called mal-operation of the relays.



Generator G1 feeding parallel T/L 1 and T/L 2: R1, R2 non-directional at source end (TMS = 0.3), R3, R4 directional at far end (TMS = 0.1); source -> non-directional

Backup of R_3 is R_2 ; Backup of R_4 is R_1.



G1 feeding parallel lines with R1..R4, then R5 (directional) and R6 (non-directional) on a line to G2, and R7 on the load feeder

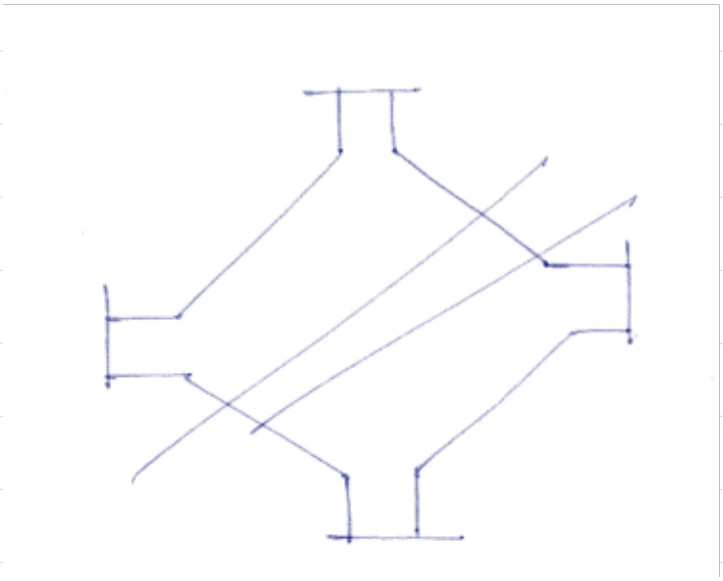
DOC / Backup

R_3 / R_2, R_6, R_7

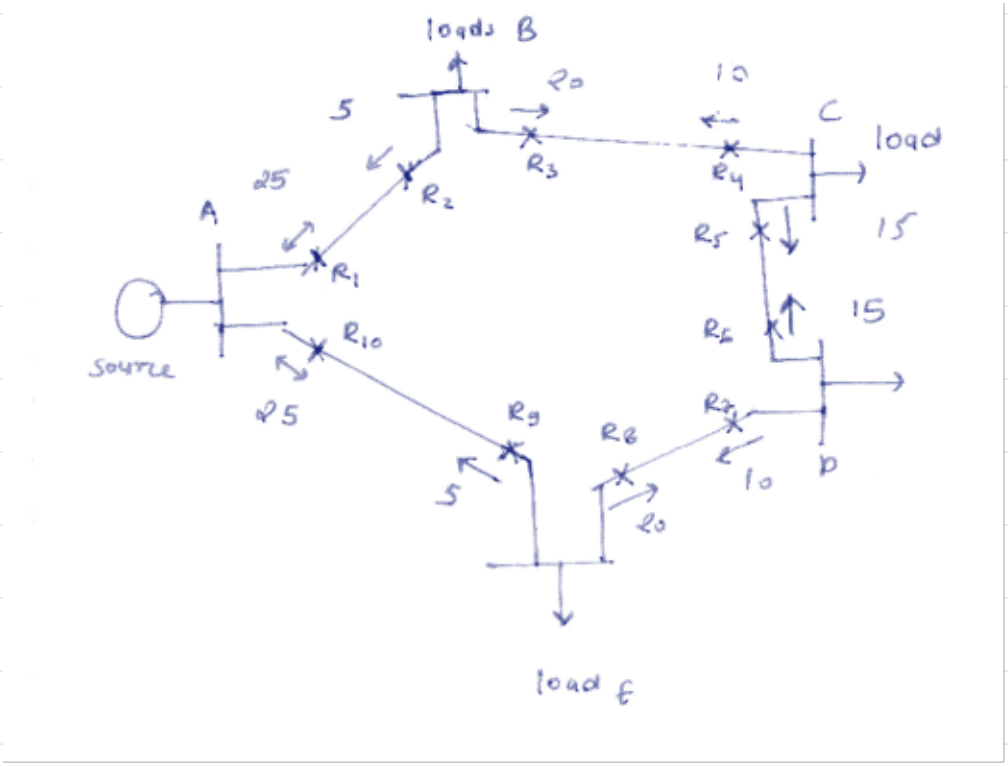
R_4 / R_1, R_6, R_7

R_5 / R_1, R_2, R_7

Protection of Ring system :

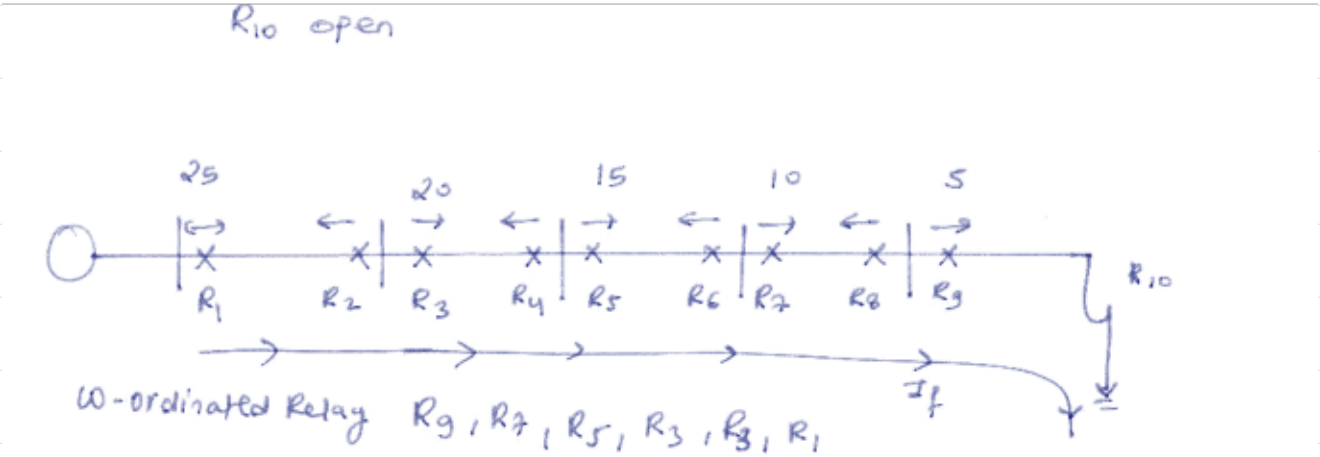


Sketch of a ring of four buses interconnected



Ring A-B-C-D-E with source at A, relays R1..R10 and directional arrows, flows 25, 5, 10, 15, 5, 10 and loads at B, C, D, E

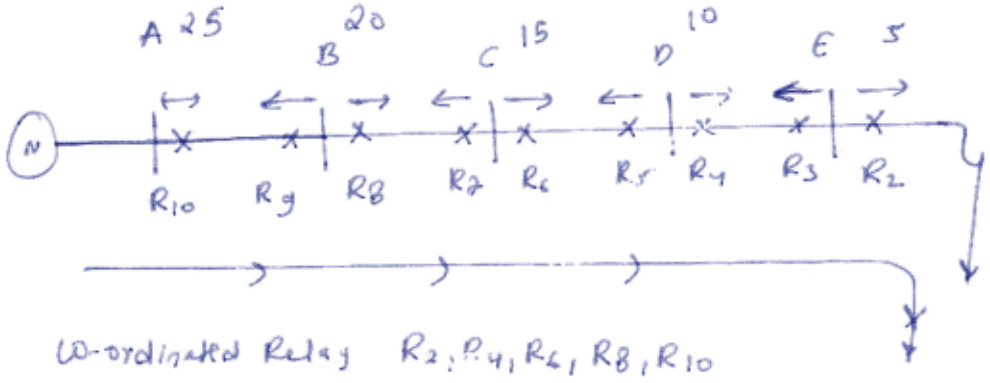
R_{10} open :



Ring cut at R10 to form a radial line from the source through R1..R9; loads/flows 25, 20, 15, 10, 5; fault at far end fed from source; co-ordinated relays R9, R7, R5, R3, R1

Co-ordinated Relays : R_9, R_7, R_5, R_3, R_1

R_1 open :



Ring cut at R1: radial line from source via A, B, C, D, E with relays R10..R2 and flows 25, 20, 15, 10, 5; co-ordinated relays R2, R4, R6, R8, R10

Co-ordinated Relays : R_2, R_4, R_6, R_8, R_{10}

Step 1 : Open one of the relays at source end and draw the radial line.

Step 2 : Create fault at far end of source and feed fault current from source, and observe relay direction and fault direction are same so that the relays are co-ordinated, providing time grading TMS from last to source end relay in increasing order.

Unit and Non unit protection Relays.

* Unit protection Relays : used to identify fault within a small physical space or unit. Ex: Differential, Buchholz, overflux Relay.
* Non unit protection Relays : used to identify faults within its own unit and also in adjacent unit faults. Ex: OC, DOC, Distance Relays.
* Through fault : Fault occurring outside unit protection relay area is called through fault.
* Mal operation of Relay : If any relay produces trip signal without fault, it is called mal operated Relay.

Overflux Relay :

* Overflux relay is used to protect the X'mer core from saturation of core.

$V = 4.44 f \Phi_m N; \quad (\Phi_m)_n \propto \left(\frac{V}{f}\right)_n$

* Trip : $\left(\frac{V}{f}\right)_f > K_1 \left(\frac{V}{f}\right)_n$ for 120 sec.
* Block : $\left(\frac{V}{f}\right)_f \leq K_1 \left(\frac{V}{f}\right)_n$; K_1 = 1.1 to 1.3 ; n = nominal , f = fault.

Normal : $\left(\frac{V}{f}\right)_f = \frac{110}{50} = 2.2$

Alarm : $\left(\frac{V}{f}\right)_f$ = 2.42 , 5 sec time delay (110 % of 2.2)

Trip : $\left(\frac{V}{f}\right)_f$ > 3.08 , 0.1 sec (140 % of 2.2)

Buchholz Relay :

* Gas operated Relay
* If any internal fault occurs in the X'mer, non flammable gases are produced passing through the Buchholz relay to the conservator. These gases produce pressure on the mercury switch inside the Buchholz relay and hence produce trip and alarm signals.
* By collecting samples of gas and oil from Buchholz relay, Dissolved Gas Analysis test is conducted, and from the % of hydrocarbons present, location is identified and repairing is done on that part.
* For protection of X'mer from internal or incipient faults.
* Its operating time is more and hence associated with % Differential Relay for faster isolation of faulty section.

Difference or Differential Relay :

Principle : (1) If a phasor or vector difference of 2 electrical quantities is exceeding the relay internal set values, it produces trip signals.

Application : used for protection of (1) Generator winding (2) X'mer winding (3) Busbar (4) T/Line

Types of Differential Relay : (1) Ordinary Diff Relay (not practically used) (2) % Differential Relay (most practically used)

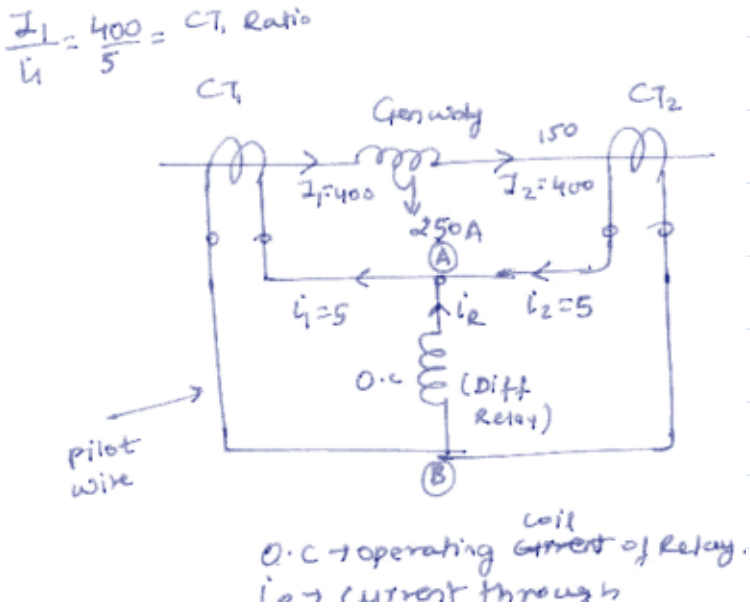
(1) Ordinary Diff. Relay :

(a) No fault - Block : $I_1 = 400$, $I_2 = 400$ (Gen wdg) ; $I_{Diff} = I_1 - I_2 = 400 - 400 = 0$

(b) Fault - Trip : $I_1 = 400$, $I_2 = 150$ A, 250 A to fault ;

$I_{Diff} = I_1 - I_2 = 400 - 150 = 250 \neq 0$

$\frac{I_1}{i_1} = \frac{400}{5} = CT_1$ Ratio ; $\frac{400}{5} = \frac{I_2}{i_2} = CT_2$ Ratio



Generator winding between CT1 and CT2 (I1 = 400, I2 = 400, 150 on fault), pilot wires to O.C operating coil of differential relay between nodes A and B; i1 = 5, i2 = 5, relay current iR

O.C -> operating coil of Relay ; i_R -> current through relay coil.

Node A : $i_R + i_2 = i_1$ -> $i_R = i_1 - i_2$

$$i_R = \frac{I_1}{CT_1 \text{ Ratio}} - \frac{I_2}{CT_2 \text{ Ratio}}$$

$CT_1 = CT_2 = CT$ Ratio :

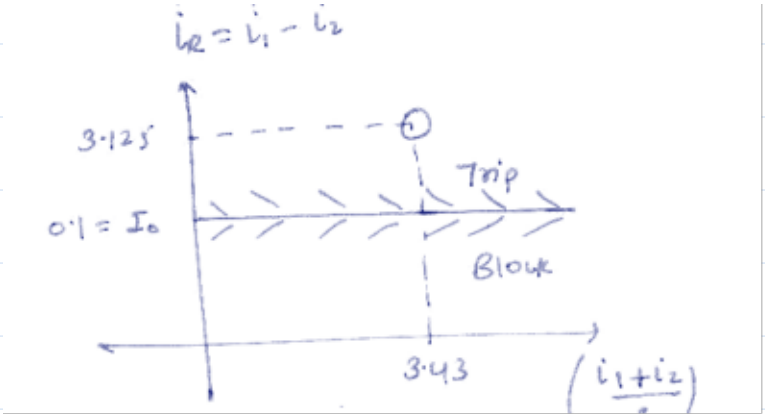
$$i_R = \frac{I_1 - I_2}{CT \text{ Ratio}}$$

Fault : $i_R = \frac{400-150}{400/5} = 3.125\ A$; assume $I_o = 0.1\ A$; $i_1 = 5$; $i_2 = \frac{150}{(400/5)} = 1.875\ A$; $\frac{i_1+i_2}{2} = 3.43\ A$

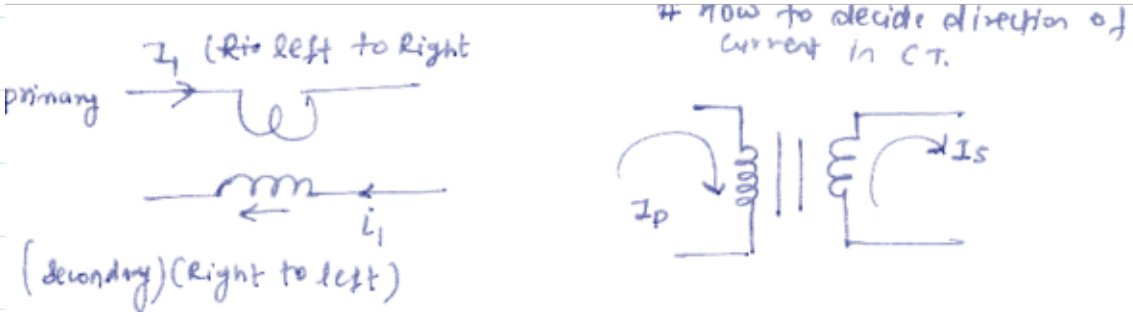
Block : $i_R = 0$ (Ideal) ; $i_R \leq I_o$ (practical)

Trip : $i_R \neq 0$ (Ideal) ; $i_R > I_o$ (practical)

I_o = min pickup current required to overcome spring restraining torque



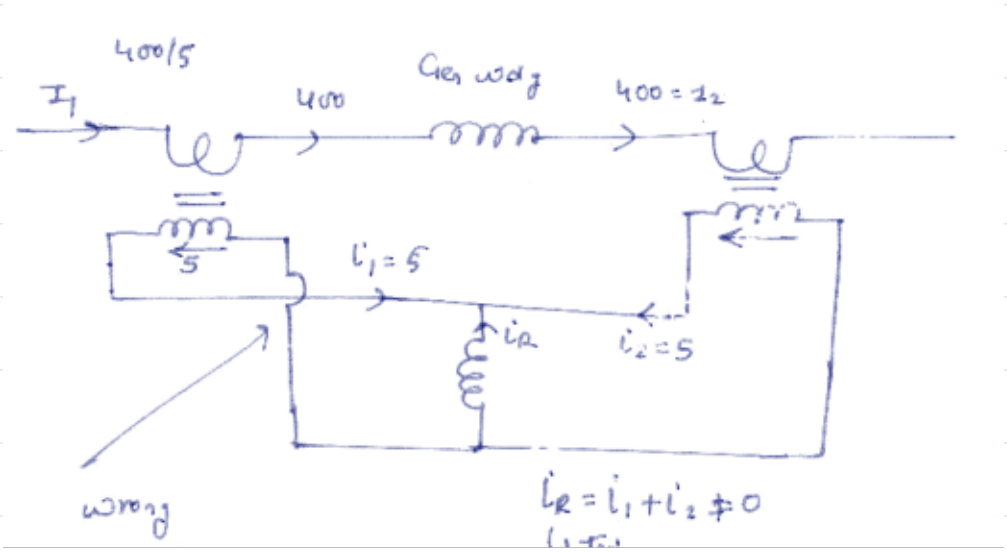
Plot of iR = i1 - i2 against (i1 + i2)/2: trip above the line at I_o = 0.1, block below; fault point at 3.125 A and 3.43 A



CT primary current I1 flows left to right, CT secondary current i1 flows right to left; how to decide direction of current in CT: Ip and Is

Disadvantages of ordinary Differential Relay :

(1) Wrong connection in the pilot wire at one of the CT side causes $I_R \neq 0$ so that Relay mal-operates.

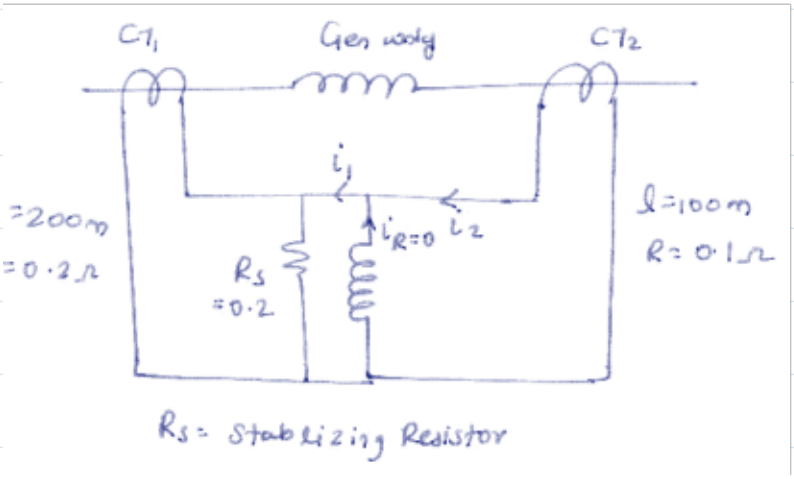


Generator winding with 400/5 CTs; one CT secondary wrongly connected so i1 = 5 and i2 = 5 add: iR = i1 + i2 not equal to 0, relay trips

(2) Difference in length or resistance of pilot wire causes unequal current flowing through the pilot wire so that relay mal-operates.

* To compensate these effects, stabilizing resistor is connected across the relay coil.

* Stabilising resistor is connected parallel to relay operating coil to compensate resistance effect of pilot wire.



Generator winding between CT1 and CT2 with pilot wires of 200 m (0.2 ohm) and 100 m (0.1 ohm); relay coil with stabilizing resistor Rs = 0.2 ohm in series; iR = 0

R_s = Stabilizing Resistor

(3) If one of CT core is saturated then Relay mal-operates.

(4) Due to Ratio and phase angle errors, unequal current flows through pilot wire so that Relay mal-operates.

I_f | CT1 | CT2 | $i_R = i_1 - i_2$

200 | 200/2 | 200/2 | 2 - 2 = 0 (Block)

400 | 400/4 | 400/4 | 4 - 4 = 0 (Block)

600 | 600/6 | 600/4 | 6 - 4 = 2 ≠ 0 (Trip)

800 | 800/8 | 800/4 | 8 - 4 = 4 ≠ 0 (Trip)

* Ideal X'mer: Ip and Is have phase angle difference of 180 degrees. But due to Io this difference is less than 180 degrees. This will produce phase angle error.

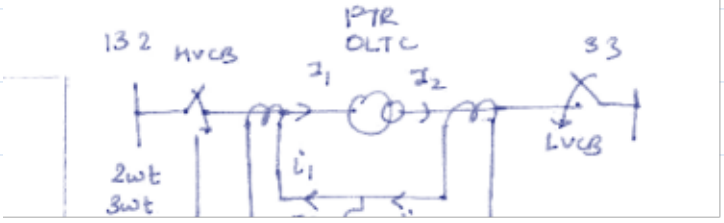
DAY # 12 [20/9/25]

(5) During changing of OLTC tap of the Power X'mer, unequal current flows through the pilot wire which causes mal-operation of the relays.

(6) During initial charging time of the power X'mer, huge amount of current called magnetising inrush current flows through the X'mer, which will contain 2nd, 3rd, 4th etc. harmonics and causes unequal current to flow through the pilot wire and hence relay mal-operates. To overcome this effect an additional Relay called Harmonic Restraint

Relay is used which will observe 2nd, 3rd, 4th etc. harmonics, produce additional restraining torque and block the relay mal-operation during initial charging time. After normal condition is restored, HRR will unblock the differential relay and relay is ready for service for identification of faulty conditions.

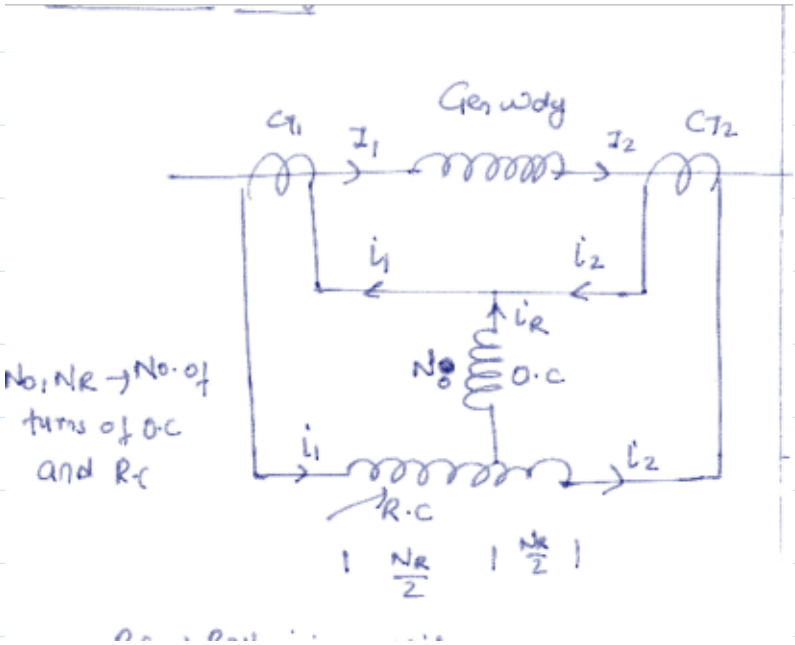
d 2nd, 3rd, 4th etc. harmo
Torque and block the relay
ime. After normal condition
fferential Relay and relay is
faulty conditions.



132 kV HV CB, power transformer with OLTC (I1, I2), 33 kV LV CB; differential relay with trip coil and harmonic restraint relay (HRR) fed with 2wt, 3wt, 4wt components

$I_1 = I_{m1}\sin(\omega t) + I_{m2}\sin(2\omega t) + I_{m3}\sin(3\omega t) + \cdots$; HRR = harmonic Restraint Relay

% Differential Relay :



$N_o, N_R \rightarrow$ No. of turns of O.C and R.C

Generator winding with CT1, CT2 and currents I1, I2; restraining coil R.C (Nr/2 turns each half) and operating coil O.C (No turns) with iR, i1, i2; No, Nr = number of turns of O.C and R.C

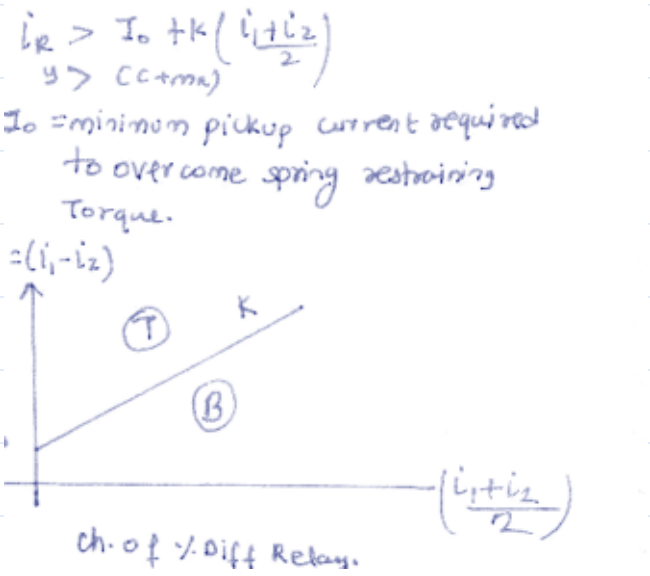
R.C -> Restraining coil ; O.C -> operating coil.

Operating mmf = $N_o \cdot i_R$; Restraining mmf = $i_1\frac{N_R}{2} + i_2\frac{N_R}{2} = N_R\left[\frac{i_1+i_2}{2}\right]$

$N_o i_R = N_R\left[\frac{i_1+i_2}{2}\right]$; $K = \frac{i_R}{\left(\frac{i_1+i_2}{2}\right)} = \frac{N_R}{N_o}$; K = Slope or % Bias

(1) Block : $i_R \leq \left[I_o + K\left(\frac{i_1+i_2}{2}\right)\right]$

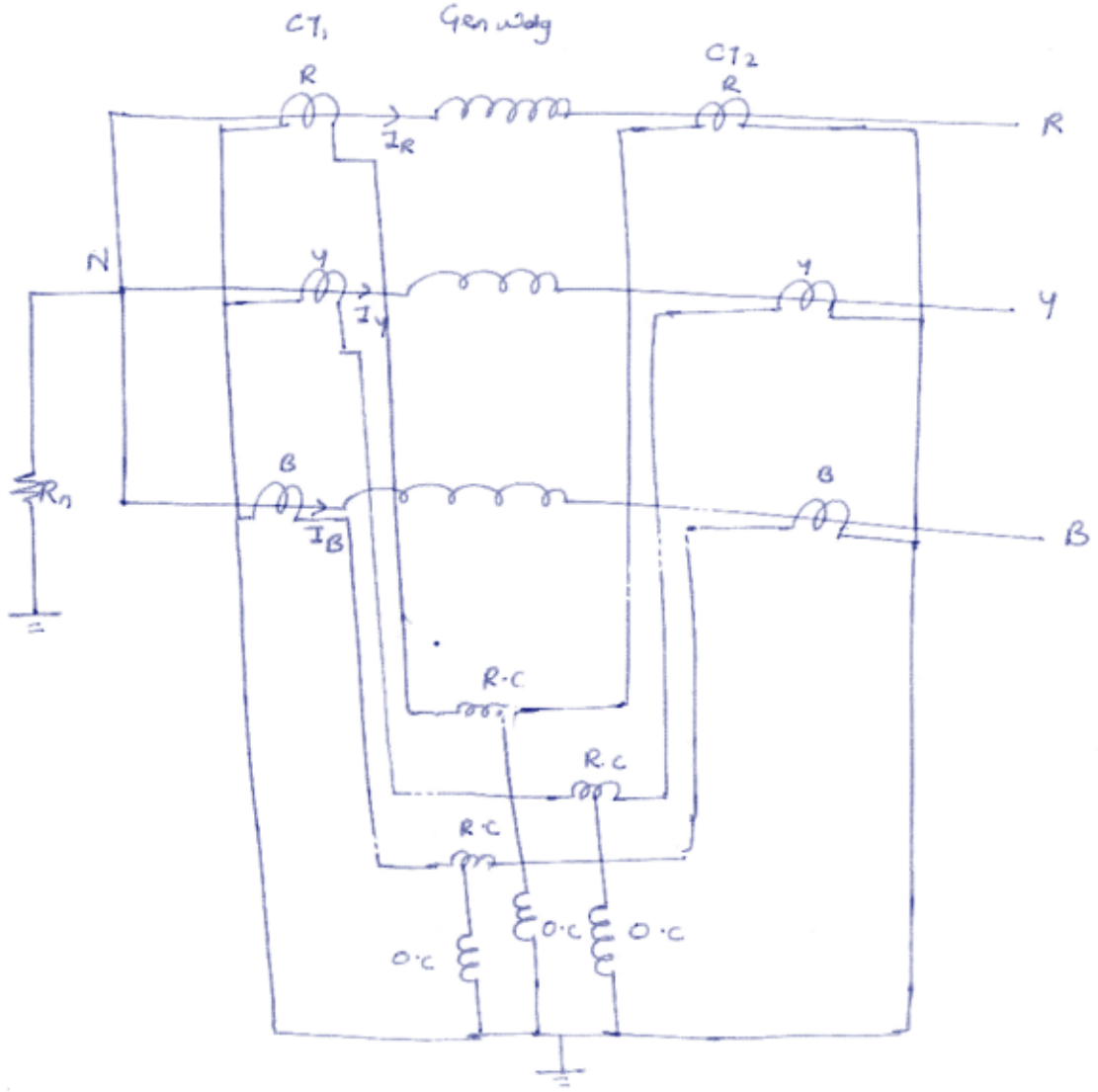
(2) Trip : $i_R > I_o + K\left(\frac{i_1 + i_2}{2}\right)$; I_o = minimum pickup current required to overcome spring restraining torque.



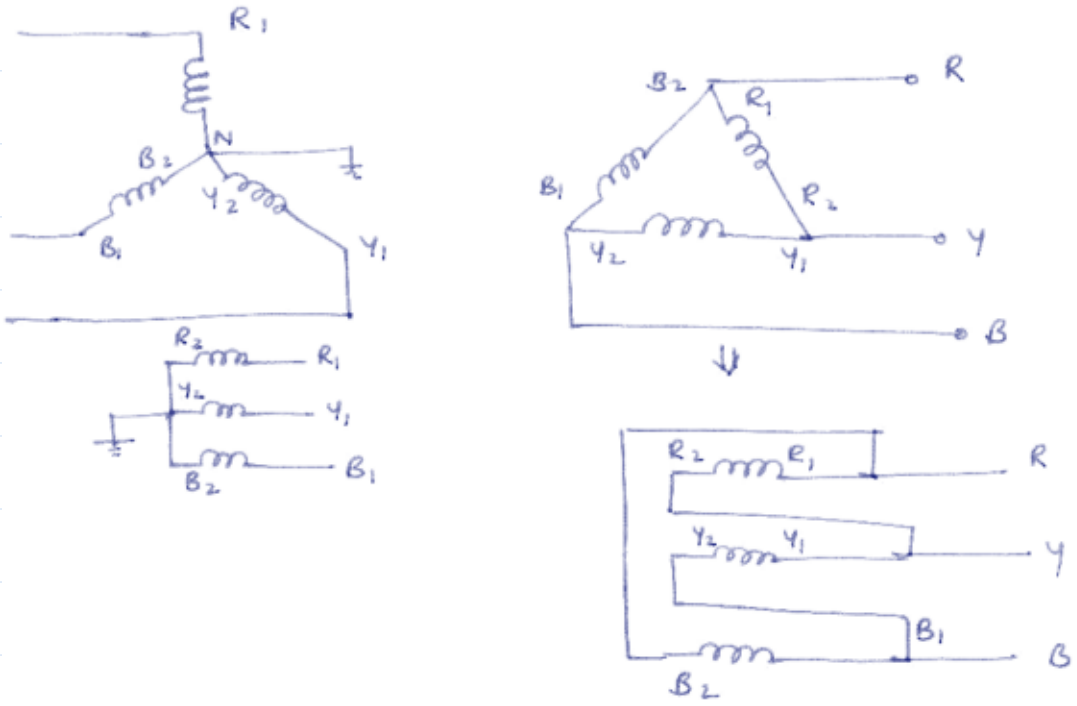
Characteristic of % diff relay: iR = (i1 - i2) against (i1 + i2)/2, line of slope K starting at I0; trip (T) above, block (B) below

% Diff. Relay is most practically used to overcome the disadvantage of ordinary Diff. relay.

(1) Protection of 3-ϕ Gen. winding using % Diff. relay.



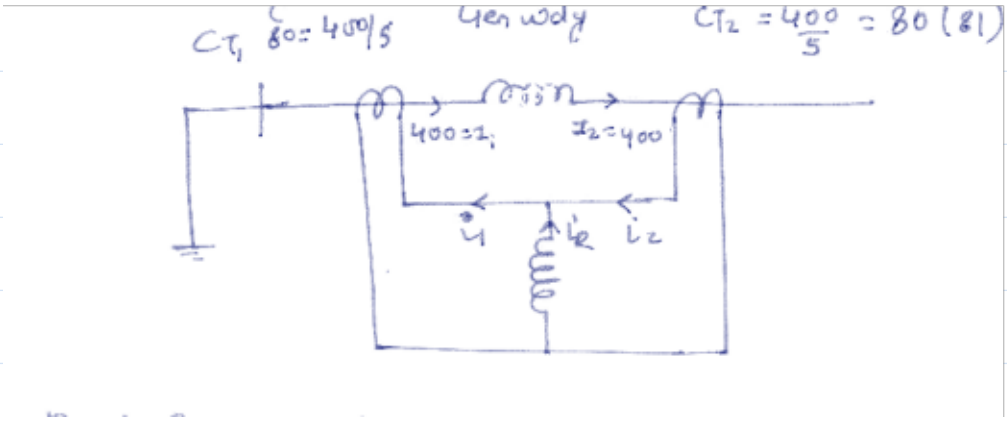
Three-phase generator windings R, Y, B with neutral N earthed through Rn; CT1 and CT2 on each phase; three restraining coils R.C and three operating coils O.C connected through pilot wires



Generator winding arrangements: star with split windings R1/R2, Y1/Y2, B1/B2 and neutral N; delta with double windings arranged so that both halves of each phase are in the differential zone

W.B Q.24

(b) Ordinary Diff : CT1 = 400/5 = 80 (79) ; CT2 = 400/5 = 80 (81) ; $I_1 = I_2 = 400$



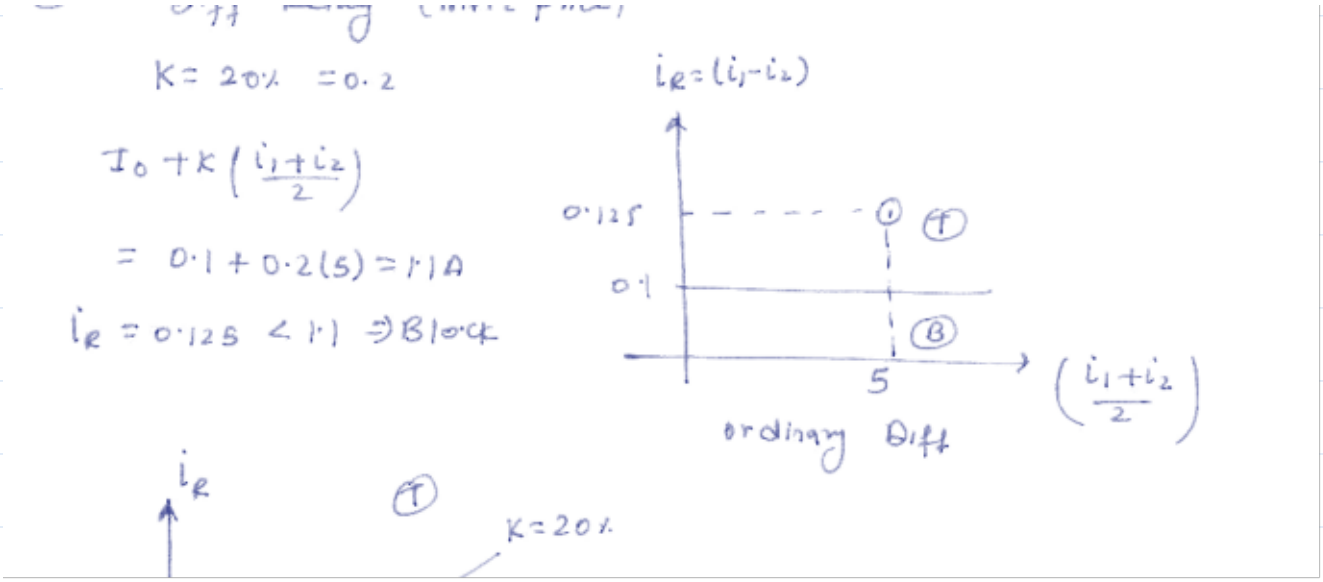
Generator winding with CT1 and CT2 and relay coil: i1, iR, i2

Due to ratio error : $i_1 = \frac{400}{79} = 5.063$; $i_2 = \frac{400}{81} = 4.938$; $\frac{i_1 + i_2}{2} = 5\ A$

$i_R = i_1 - i_2 = 0.125 > I_o$ -> Trip

(c) % Diff Relay (more price) : K = 20 % = 0.2

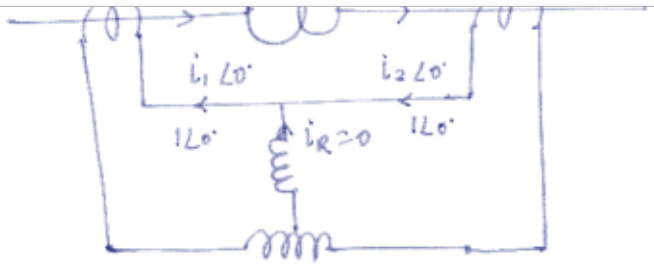
$I_o + K\left(\frac{i_1 + i_2}{2}\right) = 0.1 + 0.2(5) = 1.1\ A$; $i_R = 0.125 < 1.1$ => Block



Ordinary differential: iR = (i1 - i2) against (i1 + i2)/2 shows 0.125 above I0 = 0.1: trip; % differential with K = 20 %: iR = 0.125 well below 1.1 at 5 A: block

Protection of 3-ϕ Power Transformer winding using percentage Diff Relay :

1. In case 3-ϕ X'mer of Star-Delta and Delta-Star configuration, there exists a 30 degree phase displacement between primary and secondary; this will cause unequal current flow through the pilot wire so that relay mal-operates.
2. To compensate this effect, CT on corresponding sides of power X'mer are connected in opposite configurations of the PTR connection.
3. In Numerical Relays, type of X'mer data is entered into the relay through software; it will automatically compensate 30 degree phase shift.



Delta-star power transformer with CT1 (delta) and star-connected CT on other side; I1 angle 0, I2 angle 30 on the transformer side, i1 and i2 both angle 0 after CT connection compensation, iR = 0

Power X'mer connection vs CT connection :

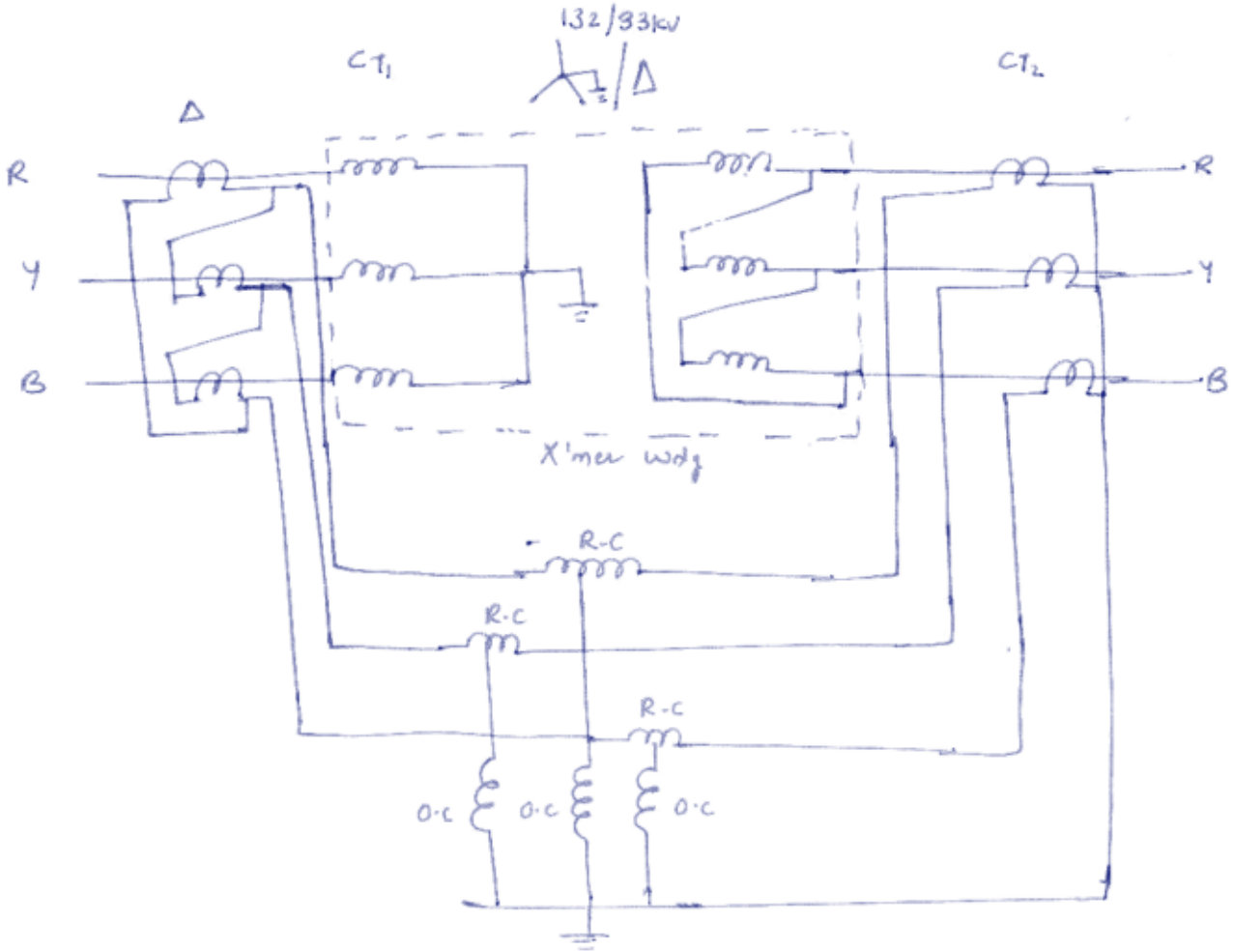
Primary / Secondary / CT1 / CT2

Delta / Star / Star / Delta

Star / Delta / Delta / Star

Star / Star / Delta / Delta

Delta / Delta / Star / Star

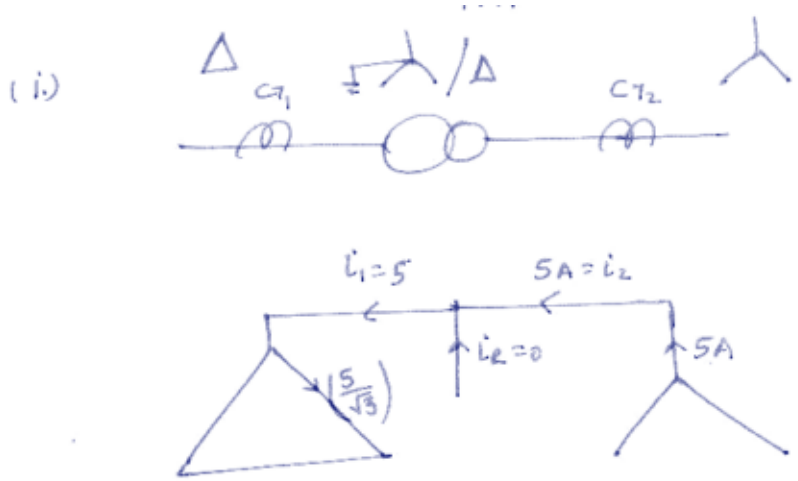


132/33 kV star-delta power transformer winding with CT1 on the delta side and CT2 on the star side connected in opposite configurations; three restraining coils R.C and three operating coils O.C connected through pilot wires; phases R, Y, B on both sides

A 132/33 kV PTR has 800/5 CT on LV side. Find the CT Ratio to be used on HV side for the following config.

(i) Star-Delta (ii) Delta-Star

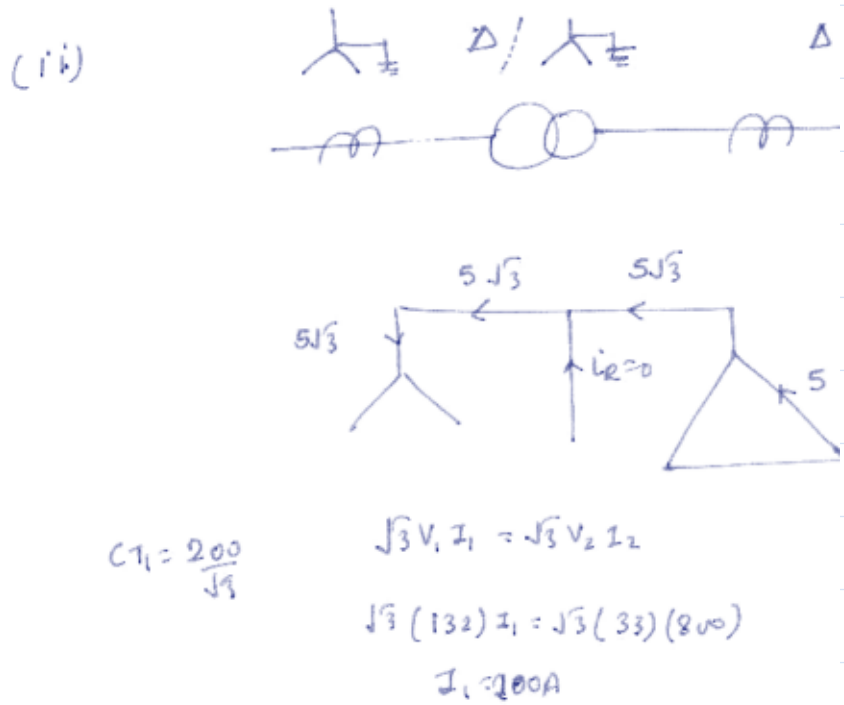
(i) 132/33 kV star / delta : CT1 on the star side is connected in delta; CT2 on the delta side in star.



Transformer with CT1 (delta connected) and CT2 (star connected); currents i1 = 5 A, i2 = 5 A; each delta winding carries 5/sqrt3; iR = 0

$$CT_1 = \frac{200}{5/\sqrt{3}} \; ; \sqrt{3} V_1 I_1 = \sqrt{3} V_2 I_2 \, , \sqrt{3}(132)(I_1) = \sqrt{3}(33)(800) \rightarrow I_1 = 200 \text{ A}$$

(ii) Delta / star : CT1 on the delta side in star, CT2 on the star side in delta.



$CT_1 = \frac{200}{\sqrt{3}}$

$\sqrt{3} V_1 I_1 = \sqrt{3} V_2 I_2$

$\sqrt{3}(132) I_1 = \sqrt{3}(33)(800)$

$I_1 = 200A$

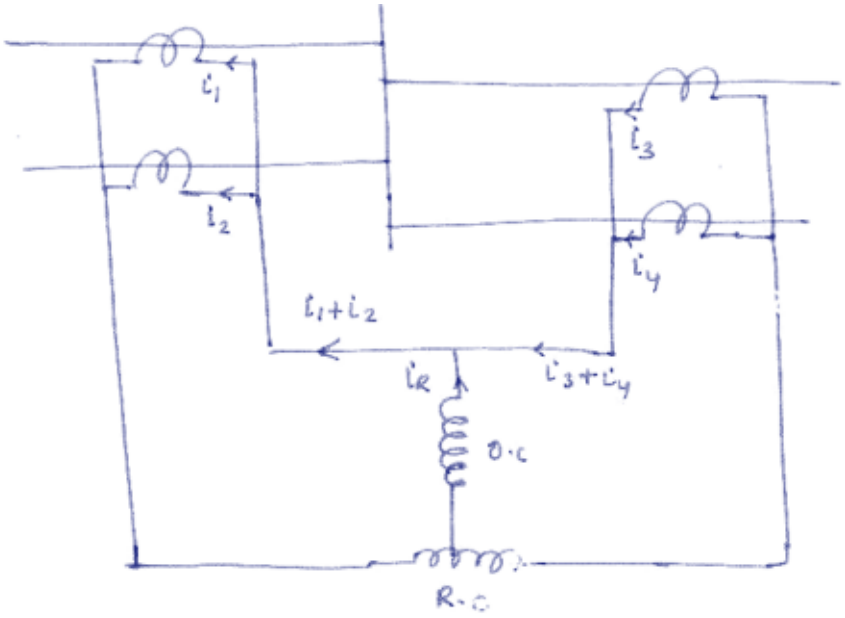
Transformer D/Y with CT in star on HV and CT in delta on LV; currents 5 sqrt3 on star side, 5 on each delta winding; iR = 0

$$CT_1 = \frac{200}{\sqrt{3}} \; ; \sqrt{3}(132) I_1 = \sqrt{3}(33)(800) \rightarrow I_1 = 200 \text{ A}$$

Protection of Busbars using percentage Differential Relay :

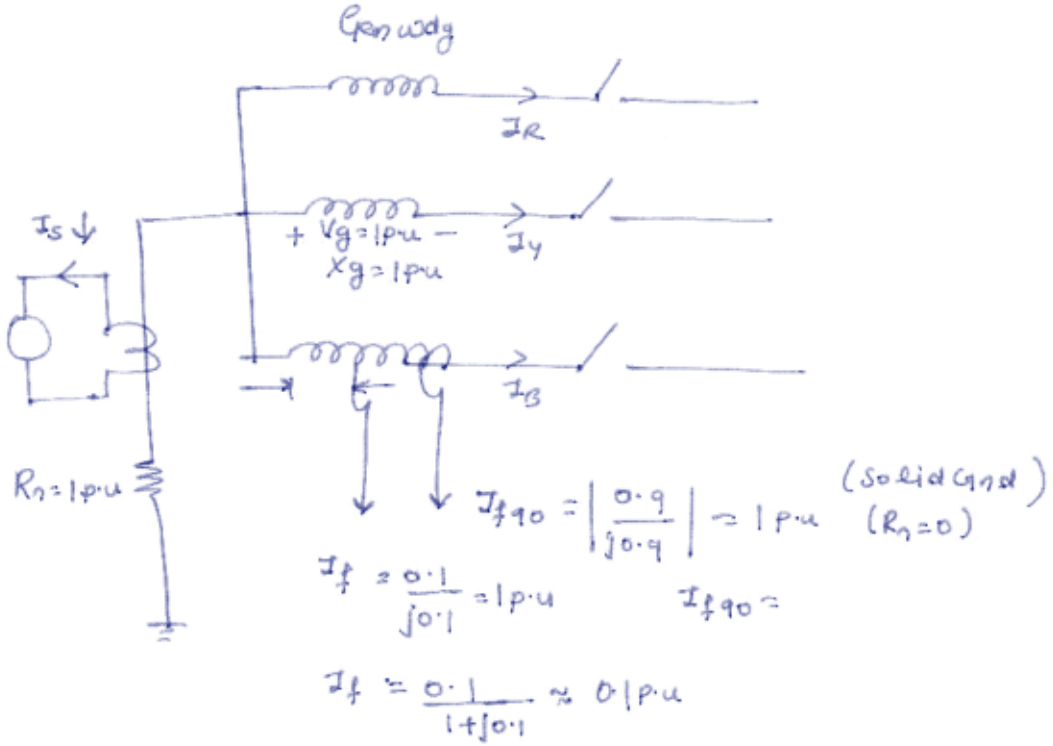
In modern S/S, 2 bus double bus system and 1 1/2 bus system are practically used to improve the system stability.

If any fault occurs on one of the buses, correspondingly connected all the feeder breakers are isolated to improve the reliability.



Bus with four feeder CTs: i1, i2 on one side and i3, i4 on the other; currents (i1 + i2) and (i3 + i4) summed into operating coil O.C (iR) and restraining coil R.C

Restricted Earth fault Relay : [Generator]



Generator winding R, Y, B with CTs on each phase, neutral earthed through Rn = 1 p.u with CT and relay (Is); Vg = 1 p.u, Xg = 1 p.u; fault in winding of phase B at 10 % and 90 % points

$$I_{f90} = \left|\frac{0.9}{j0.9}\right| = 1 \text{ p.u (solid Gnd, } R_n = 0); \quad I_f = \frac{0.1}{j0.1} = 1 \text{ p.u}; \quad I_f = \frac{0.1}{1+j0.1} \approx 0.1 \text{ p.u}$$

Restricted E/F Relay formulae :

$$\% \text{ winding unprotected} = \%W_u = \frac{I_n R_n}{V_{ph}} \times 100$$

$V_{ph} = \frac{V_L}{\sqrt{3}}$; R_n = Neutral Resistance

Case 1 : CT Ratio is given : I_n = Minimum pick up current of Relay x CT Ratio = out of balance current value in Amps x CT Ratio

Case 2 : CT Ratio is not given and out of balance current is given : I_n = % of out of balance current x Gen full load current (I_g) ; $I_g = \frac{S_{3\phi}}{\sqrt{3} V_L}$

% winding protected = $\% W_p = 100 - \% W_u$

Eg: I_n = Minimum pick up current of Relay x CT Ratio = $0.75 \times \frac{1000}{5}$

$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{11 \times 10^3}{\sqrt{3}}$; $\%W_u = \frac{I_n R_n}{V_{ph}} \times 100 = 11.8\%$; $\%W_p = 100 - \%W_u = 88.2\%$

Eg: An 11 kV / 100 MVA alternator is protected with % diff relay. The % of winding to be protected against phase to Gnd fault is 85 %. The relay is set to operate when there is 20 % out of balance current. Determine the value of resistance to be placed in the Neutral to Gnd connection ?

Case 2 : $I_g = \frac{S_{3\phi}}{\sqrt{3}\,V_L} = \frac{100 \times 10^6}{\sqrt{3} \times 11 \times 10^3} = 5.25\ kA$; $V_{ph} = \frac{11 \times 10^3}{\sqrt{3}} = 6.35\ kV$

I_n = out of balance current x Gen full load current (I_g) = 0.2 x 5.25 = 1.05 kA

$\%W_u = 100 - 85 = 15\ \% = \frac{I_n R_n}{V_{ph}} \times 100 = \frac{1.05 R_n}{6.35} \times 100 \rightarrow R_n = 0.907\ \Omega$

3-ϕ X'mer protection Relays :

HV side : O/C , Over vtg , E/F ; LV side : O/C , Over vtg , E/F

Common Relays both HV and LV side : % Differential Relay , Over flux , Buchholz Relay.

3-ϕ Generator protection relays :

Fault -> Name of Relay

-> Stator faults

* Phase-phase -> % Differential
* Inter-phase -> Split phase
* E/F -> REF (Restricted E/F)

-> Rotor faults

* E/F -> Sensitive Rotor E/F

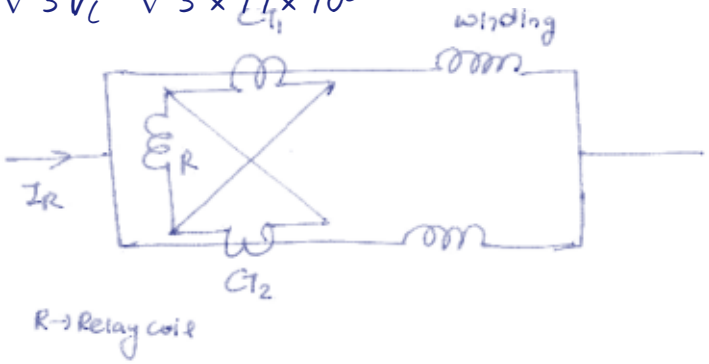
-> External faults

* Failure of prime mover -> Reverse power
* Failure of Excitation -> Offset MHO Relay
* Failure of cooling medium -> Thermal / Temp sensitive Relay
* Over speed -> Fly ball Governor (Mechanical Centrifugal Device)
* Unbalanced loading -> Negative sequence current Relay.

* Stability or load Angle -> Pole slip Relay.

Inter phase / Inter turn fault / Split phase Relay.

$$I_g = \frac{S_{3\phi}}{\sqrt{3}\,V_L} = \frac{100 \times 10^6}{\sqrt{3} \times 11 \times 10^3} = 5.25\ kA$$



Split-phase protection: generator winding split into two parallel halves, CT1 and CT2 on each half, relay coil R across them; IR the input current

R -> Relay coil

Universal Torque Eqn :

$$T_d = K_1 I^2 + K_2 V^2 + K_3 VI\cos(\theta - \tau) - K_4$$

$K_1 I^2$ - O/C , $K_2 V^2$ - O/V , $K_3 VI\cos(\theta - \tau)$ - Directional ; K_4 - spring Restraining Torque , T_d - operating Torque

(1) O/C Relay : $K_2 = K_3 = 0$; $T_d = K_1 I^2 - K_4$

Trip : $T_d > 0$; $K_1 I^2 > K_4 \Rightarrow I > \sqrt{\frac{K_4}{K_1}}$; I = Fault current ; $\sqrt{K_4/K_1}$ = % PSB setting.

Block : $T_d \leq 0$; $I \leq \sqrt{\frac{K_4}{K_1}}$

(2) O/V Relay : $K_1 = 0$, $K_3 = 0$; $T_d = K_2 V^2 - K_4$

Trip : $V > \sqrt{\frac{K_4}{K_2}}$; Block : $V \leq \sqrt{\frac{K_4}{K_2}}$

(3) Directional : $K_2 = K_1 = 0$; $T_d = K_3 VI\cos(\theta - \tau) - K_4$

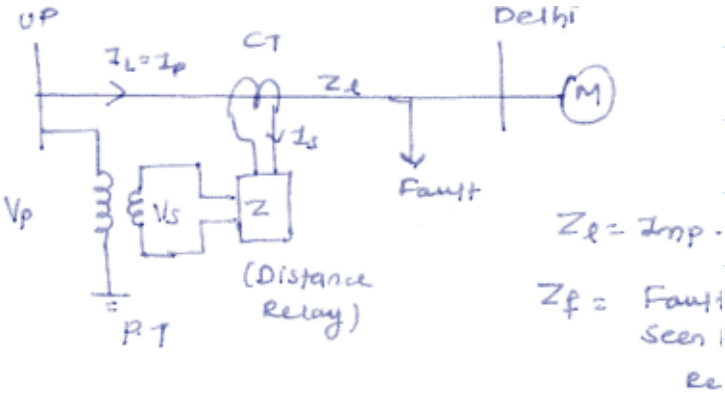
Trip : $VI\cos(\theta - \tau) > \left(\frac{K_4}{K_3}\right)$; Block : $VI\cos(\theta - \tau) \leq \left(\frac{K_4}{K_3}\right)$

Protection of T/L using distance Relays :

* Impedance of T/L is proportional to length of the line or distance of the line.

* By measuring the impedance, distance of fault location is measured.

* This relay is practically used in 132 kV and above vtg level. In modern protection systems, multiple distance relays are used in each line for improvement of system stability.



UP bus to Delhi bus line with CT (Is) and PT (Vs) feeding the distance relay Z; line impedance Zl; fault location on the line; motor load

Z_l = Imp. of T/L = $(R + jX_l)$; Z_f = Fault Impedance seen by distance Relay = $\frac{V_s}{I_s}$

(I) No Fault - Block : (a) Peak load time : assume V_s = 1 p.u, I_s = 1 p.u ; Z_l = 0.6 + j0.8 = 1 p.u ; $Z_f = \frac{V_s}{I_s} = \frac{1}{1}$ = 1 p.u = Z_l

(b) Off peak load time : assume I_s = 0.5 p.u ; $Z_f = \frac{1}{0.5}$ = 2 p.u > Z_l

(II) Fault - Trip : assume during fault I_s = 2 p.u ; $Z_f = \frac{1}{2}$ = 0.5 p.u < Z_l

Trip : $Z_f < Z_l$; Block : $Z_f \geq Z_l$

Types of Distance Relay :

(1) Impedance (Z) Relay : [Vtg Restraint O/C Relay]

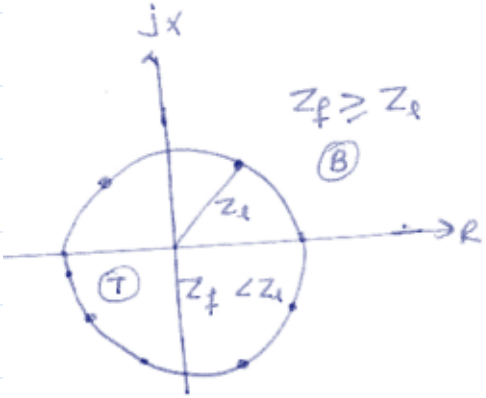
Assume UTE : $K_3 = K_4 = 0$, K_2 = -ve ; $T_d = K_1 I^2 - K_2 V^2$ (O/C and Vtg Restraint)

Trip : $T_d > 0$; $K_1 I^2 > K_2 V^2$; $\sqrt{\frac{K_1}{K_2}} > \frac{V}{I}$; $Z_l = \sqrt{\frac{K_1}{K_2}}$, $Z_f = \frac{V}{I}$

$Z_l > Z_f$ or $Z_f < Z_l$; Block : $Z_f \geq Z_l$

Locus of Impedance Relay :

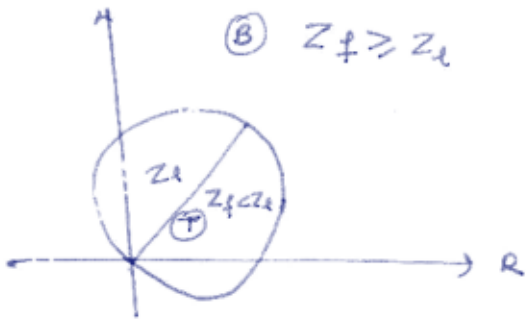
$Z_1 = R_1 + jX_1$; $Z_1^2 = R_1^2 + X_1^2$; $r^2 = x^2 + y^2$ circle eqn. Locus of Z Relay is a circle with $r = |Z_1|$



R-X diagram: circle of radius Z1 centred at origin; inside Zf < Z1 trip (T), outside Zf >= Z1 block (B)

(2) Mho (Y) / Admittance Relay : [Directional Restraint O/V Relay]

Assume $K_1 = K_4 = 0$; $T_d = K_2V^2 - K_3VI\cos(\theta - \tau)$; locus of mho relay is a circle with diameter of circle Z_1 ; $Z_f < Z_1$ - Trip ; $Z_f \geq Z_1$ - Block

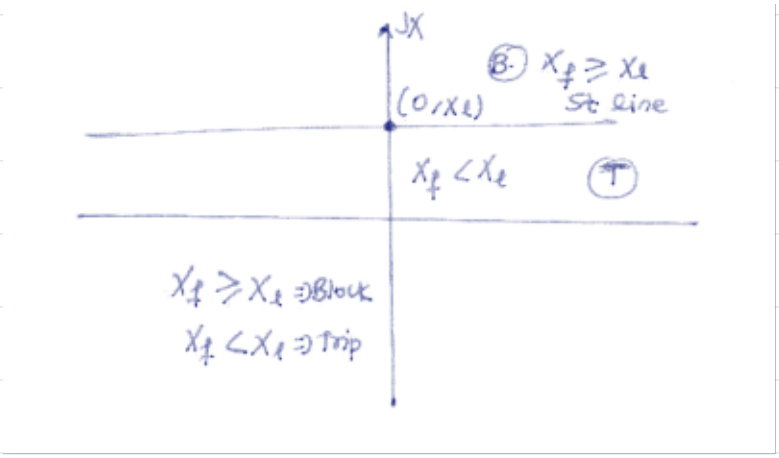


R-X diagram: mho circle through the origin with diameter Z1; inside trip, outside block

(3) Reactance (X) Relay :

Assume $K_2 = K_4 = 0$, K_1 = -ve ; $T_d = K_3VI\cos(\theta - \tau) - K_1I^2$; Current Restraint Directional Relay.

Locus of X-Relay is a straight line passing through $(0, X_1)$; $X_f \geq X_1$ => Block ; $X_f < X_1$ => Trip



R-X diagram: horizontal straight line at X = Xl; below it Xf < Xl trip (T), above it Xf >= Xl block (B)

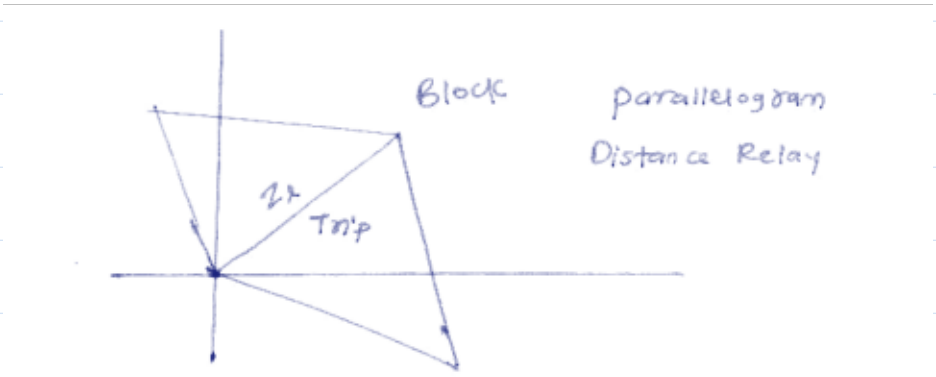
Type of line / Distance Relay used

Short / X

Medium / Z

Long / Y

X-Relay is independent of arc resistance.



Parallelogram distance relay characteristic on the R-X plane: trip (Zr) inside, block outside

Terminology :

(1) Reach : length or distance of the line protected by the distance relay is called reach of the relay.

(2) Underreach : If the distance relay is failed to identify fault within its zone or distance, it is called underreach.

(3) Overreach : If the distance relay is able to identify the fault beyond its reach, it is called overreach. Overreach is used as backup protection.

Application of overreach : 3 zone protection of T/L using distance relay.

In case of 132 kV and above vtg level, 3 zone protection is used. In this protection system each line at both ends is provided with distance relay and each distance relay internally has 3 relays i.e. Z_1, Z_2, Z_3.

Zone 1 (Z_1) :

* It covers 80 % of length of line to be protected and operating time is zero.

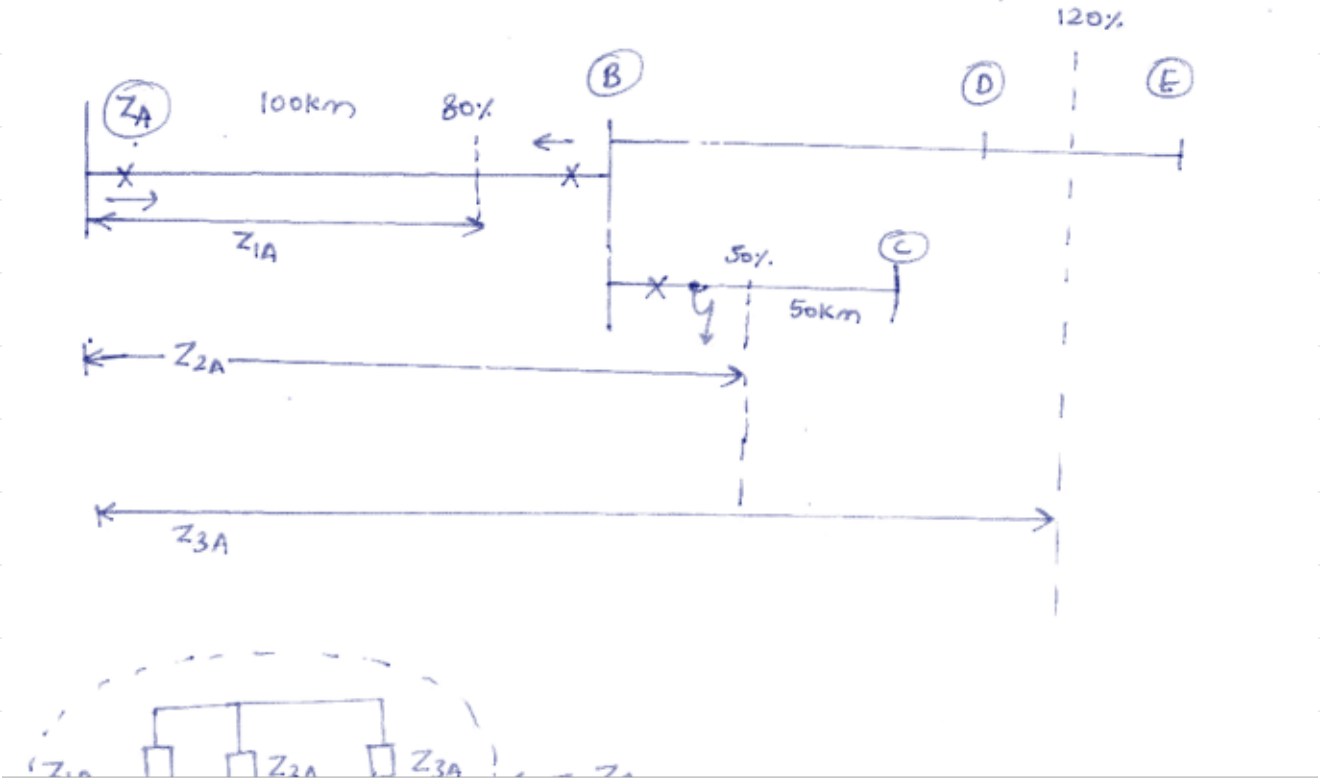
Zone 2 (Z_2) :

It covers 100 % of the 1st line + 50 % of 2nd shortest line. Operating time 0.3 to 0.5 sec.

Zone 3 (Z_3) :

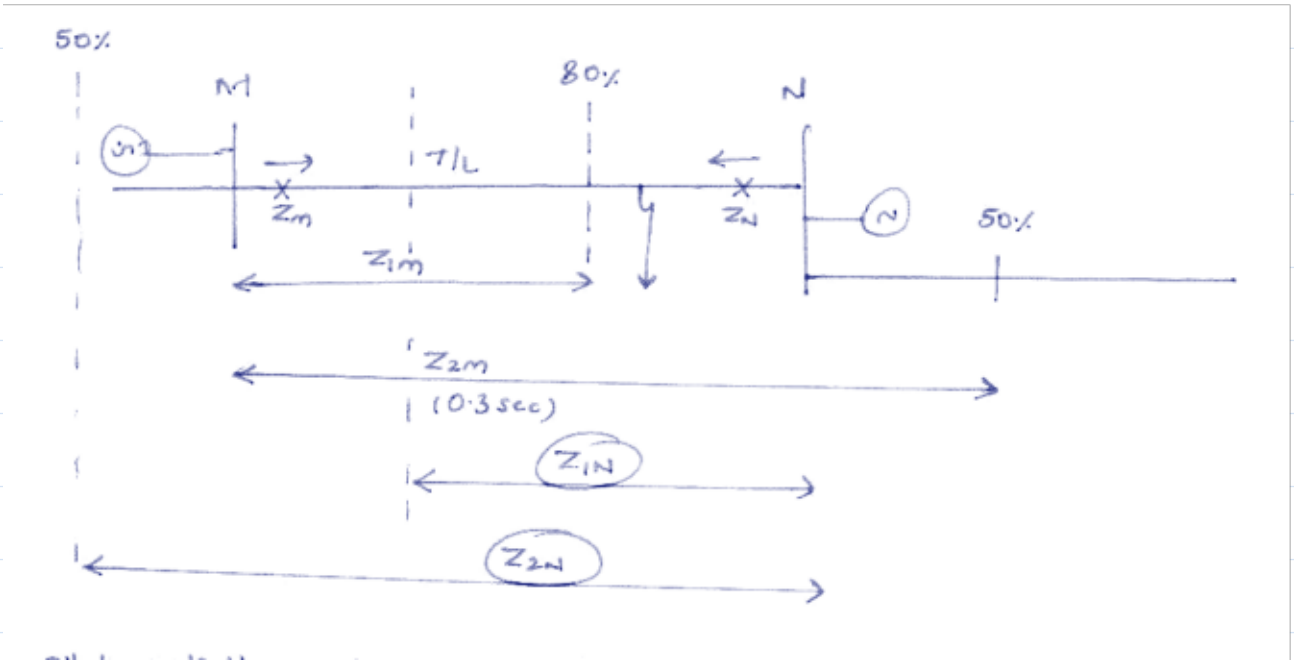
It protects 100 % of the 1st line + 120 % of the adjacent longest line. Operating time is 0.6 to 0.9 sec.

* Z_1 and Z_3 are used as a backup protection.



Line A-B of 100 km with Z1A covering 80 %, Z2A reaching 50 % of the shorter adjacent line B-C (50 km), Z3A reaching 120 % of the longest adjacent line B-E; below, relay at A made of Z1A, Z2A, Z3A elements in parallel

Pilot or Carrier protection system :



Line M-N with distance relays Zm at M and ZN at N: Z1M covers 80 % and Z2M (0.3 sec) reaches 50 % beyond N; Z1N and Z2N from the other end; sources at both ends

Pilot protection system is used to improve system stability by reducing the time of operation of Z_2 and Z_3 relays by sending a carrier signal to the other side of the S/S. This is called direct Trip signal.